

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
S Susmitha	192571068	ITA0510	Computer Vision

32. Morphological operations based on OpenCV using Closing technique.

AIM:

Morphological operations using Closing technique.

PROGRAM:

```
import cv2
import numpy as np

img = cv2.imread(
    r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png",
    cv2.IMREAD_GRAYSCALE
)

if img is None:
    print("Error: Image not found!")
else:
    # Create a 5x5 kernel
    kernel = np.ones((5, 5), np.uint8)

    # Apply Closing operation (Dilation followed by Erosion)
    closing = cv2.morphologyEx(img, cv2.MORPH_CLOSE, kernel)

    # Save the output image
    cv2.imwrite(
        r"C:\Users\Susmitha\Downloads\ITA0510 Lab\Closing.jpg",
        closing
    )

    # Display the images
    cv2.imshow("Original Image", img)
    cv2.imshow("Closing Image", closing)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.